Inside a language model

AI · Level 3 · Lesson 4



Today we will

- Describe the loop: tokens become numbers, attention gathers context, the model scores every possible next token, picks one and repeats.
- Explain the training stages: pretraining on huge amounts of text, then fine-tuning with examples and human feedback.
- Explain why a language model can be confidently wrong, and how grounding and checking help.

Remember

- Each round: tokens become numbers, attention gathers context, layers work on it, the next token is scored and picked.
- Pretraining makes it fluent; fine-tuning with examples and feedback makes it helpful.
- It predicts likely text, not checked facts: ground it in a source, then check.

Who is “it”?

1. Warm-up (5 min): Finish “The bus didn’t stop because it was…”.
2. Who is “it”? (12 min): Arrows from “it” to its clues.
3. Change one word (6 min): Watch the arrows move.
4. The loop (9 min): Tokens, numbers, attention, scores, pick.
5. Wrap-up (8 min): Grounding demo, then the exit ticket.

Prepare three “it” sentences in the pupils’ language, a printed chatbot answer that contains one wrong local fact, and the textbook page that corrects it.

Let’s discuss

1. How did you know what “it” meant? Which words did you look back at?
2. Why might a chatbot get a fact about our town wrong?
3. What source would you give a chatbot before trusting its answer?

Exit ticket

- Write the loop in five words, and one reason a language model can be confidently wrong.